microaqila - shopping list

12 V temperature controller (ESP board) - buy date 27 Aug 2026 - take the WHITE POWER BRICK with you

Final-final list

#	Buy	Why (human words)
1	Digital multimeter - 1	Tells you volts and ohms so you never have to guess what a wire is doing.
2	DC FEMALE jack with screw terminal - 2 pcs	The power brick ends in a plug with nothing to plug into. This is the socket - it turns the brick into two screw-down wires. TEST THE CLICK on the brick's plug before paying.
3	12V PTC heater pad, small (10-30 W) - 1	The heater. This type physically cannot overheat itself, even if the software dies with it stuck ON.
4	Inline fuse holder - 1, with 3A fuses x2 and 5A fuses x2	If anything shorts, a Tk-5 fuse dies instead of the wires, the brick, or the board.
5	Thick stranded wire 1 mm2 - red 1 m + black 1 m	Heater current would cook the thin jumper wires. This carries it comfortably. Cheapest thing in the shop.
6	Heat shrink packet (or electrical tape roll)	Wraps bare metal joints so nothing can touch and short.

Only if the shop has NO PTC pad

Buy instead	Why
22 ohm 10 W ceramic resistor - 4 pcs	Four together become the heater (known, bounded behaviour).
Thermal fuse 90 C - 2 pcs	Emergency brake for the resistor heater - MANDATORY with this variant (resistors don't self-limit like PTC).

NOT buying (decided)

ESP32 (the ESP8266 we have stays), current sensor (the new multimeter covers it), choc block (our screw terminals can clamp two wires each), any box/chamber (see below).

What the heck are we building?

A box-brain that holds a temperature you choose. A sensor feels the air, the ESP compares it to your target, and flips two automatic switches: too cold - heater ON; too warm - fan ON. The screen shows live numbers; later the phone sets the target. The capstone-worthy part: a dumb thermostat clicks its switch constantly right at the target line - ours keeps a smart gap around the target so it clicks far less, and we prove it with a click-counter. Relays live longer; graph goes in the report.

Demo day looks like

Power on - screen shows room temp + target. Set the target a few degrees up - heater relay clicks, temp climbs on screen, hits target, clicks off - drifts down, clicks back on. All by itself. Change the target from the phone; it obeys. Then dumb-mode vs smart-mode, show the counter: smart clicked way less. Bow.

The shoebox thing

Nothing to buy and the project is not sensitive - it's the opposite problem: a small heater warming an open room is too weak to move the number, so the demo would be everyone staring at 31.0 C forever. Any box or upturned food container placed over the sensor + heater AT DEMO TIME ONLY traps the warm air, so the number visibly climbs in a minute and the clicking show begins. It's a stage prop from the kitchen, not a component.

Say to the shopkeeper (word for word)

> "Ekta 12-volt temperature controller banachhi, ESP board diye - sensor temperature dekhe relay diye heater ar fan on/off kore. Ei list-er jinish gula lagbe."

(I'm building a 12-volt temperature controller with an ESP board - a sensor reads the temperature and relays switch a heater and fan on/off. I need the things on this list.)

If he asks anything more, two magic words cover you: "student project" - hand him the list, and let the brick do the talking for the jack size (test the click before paying).